Partial Power Conversion and High Voltage Ride-Through Scheme for a PV-Battery Based Multiport Multi-Bus Power Router

MRUDUL C.U
III YEAR EEE DEPARTMENT,
PANIMALAR ENGINEERING COLLEGE,
CHENNAI, INDIA
mrudulchaithanya2001@gmail.com

NARENDRAN M
III YEAR EEE DEPARTMENT,
PANIMALAR ENGINEERING COLLEGE,
CHENNAI, INDIA.
narennarendran012@gmail.com

K P DINAKARAN
ASST. PROFESSOR,
EEE DEPARTMENT,
PANIMALAR ENGINEERING COLLEGE,
CHENNAI, INDIA.
kpdinakaran@gmail.com

SANTHOSH KUMAR L
III YEAR EEE DEPARTMENT,
PANIMALAR ENGINEERING COLLEGI
CHENNAI, INDIA.
santhoshpec2002@gmail.com

THIRUMURUGAN A
III YEAR EEE DEPARTMENT,
PANIMALAR ENGINEERING COLLEGE,
CHENNAI, INDIA.
thirumuruganjj@gmail.com

ABSTRACT:

With the advancement of renewable technology, the distributed power supply mode, which also contains utility grid, renewable energy, and energy storage units, has gradually become a research hotspot. This paper introduces an AC/DC hybrid multi-port power routing (MPPR) system based on partial power conversion (PPC) of dual DC buses. The photovoltaic(PV)port, the battery port and two DC voltage buses for man power router. Same auxiliary port voltage modulation is used to implement PV maximum power point tracking (MPPT) and high-voltage ride through (HVRT) of the grid-tied inverter. Only the power determined by the voltage difference between the Pv modules and the series connected DC bus is dealt with in the PPC-based PV conversion, which significantly reduces the loss while compared to the full power conversion (FPC) for Photovoltaic. The detailed control schemes for fall converters and energy transmitters are provided. The simulation and experimental results verify the effectiveness of the proposed scheme.

INDEX TERMS: Partial power conversion,multi-port power routing, high voltage ride-through ,PV-battery ,grid-connected system are among the terms shown in this index.

INTRODUCTION

Renewable energy sources, such as Photovoltaic system, have rapidly developed in order to reduce use of fossil fuels & carbon emissions, and also the efficiency improvement and cost optimization of PV conversion have been studied extensively. The single stage PV power generation system has the advantages of low flow cost, compact structure, and high efficiency, however the grid connected inversion cannot be completed in low PV power conditions. Due to the features flexibility for Maximum power point and energy and control, two-stage PV conversion is also commonly used. Since it only account for a significant proportion of the total energy of the system, partial power conversion, which can be used in non-isolated conversion applications, has high efficiency and low cost features. On the basis of reference, a novel partial power conversion scheme was proposed. In a three-phase power distribution system, a conventional push-pull forward converter is used to achieve power flow management and voltage compensation. A simple high DC/DC converter suitable for medium and large-scale distributed PV applications was proposed in reference. High efficiency is achieved by means of partial
power processing as well as by coordinating the operation of

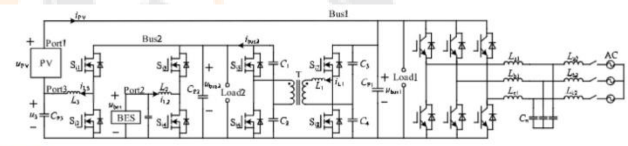

FIGURE 1. The MPPR topology of PV-battery grid-connected system.

the interleaved channels of the converter.

the ability to remove energy or smooth PV output with quick response The problem of voltage over limit and fluctuation caused by PV grid-connected can be effectively alleviated through the rational configuration of BES [16], [17]. Now, the traditional centralized power generation is gradually changing to the coexistence of centralized and distributed power generation, and unidirectional flow mode of power is gradually transforming into multidirectional modes. To improve the energy utilization of the PV-battery grid-connected system, the electrical power router has the communication and intelligent choice capabilities, which are supported by the mature application experience of the integration of micro - grid and information technology. making capabilities, and is able to realize active management of power network energy flow based on network operation state and user and control centre instructions, while also improving system power utilization effectively [18]. [19] proposed a quad-port power routing circuit that integrated storage and distributed generation, such as Photovoltaic, and enabled the implementation of power quality features. Reference[12] proposed a triple-port power routing converter integrating PV and battery power for high step-up applications, as well as a control strategy that used energy storage to maintain the Maximum power point of PV under different lighting and load conditions, realizing the multi-mode and efficient operation of the PV power generation system.

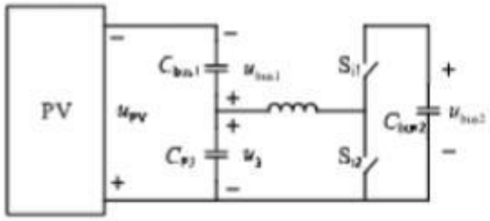

FIGURE 2. The equivalent model of PPC and HVRT.

Low-voltage ride-through strategies for PV grid-connected systems have been extensively studied, while HVRT technology is still in its infancy. SomescholarshaveconductedexploratoryresearchontheHVRTtechnologyofgrid-connected systems. To address the issue of grid voltages well leading to power backflow in the grid-side converter, proposed an HVRT control strategy with adaptive DC bus voltage adjustment in response to grid voltage changes. On the basis of an analyzing the relationship between output positive sequence reactive current and the required maximum output voltage of the grid connected converter. System based on the combination of series impedance divider and parallel high impedance grounding equipped at the wind turbine terminal, making sure the security and stability of the power grid with large-scale wind power.

An auxiliary port is designed for MPPR system in Figure 1 based on PPC to carry out MPPT of PV and HVRT of grid-connected converter by voltage modulation. A power router is made up of a PV port, a battery port, an AC port, and two DC voltage buses. The auxiliary port aids in the implementation of the PVMPPT with a PC that only transmits power determined by the electric potential difference between both the DC bus and the PV panel. The Photovoltaic panels series connected to the DC bus must be handled with in the loop, which significantly reduces the loss as compared to the FPC for PV. In the scenario where the photovoltaic conversion is unaltered, the HVRT of grid-connected converters is likewise achieved through voltage modulation. The proposed topology in this paper is introduced in next section, including the PPC and HVRT implementation methods. The operating concept of the PPC and the multi-port system's energy regulation technique were examined in depth in the third part. The simulation and experimental verification of partial power transformation and HVRT are discussed in In the last section the characteristics of the proposed topology scheme are summarized.

I. POWERROUTINGSTRUCTUREANDPARTIAL POWERCONVERSION

Figure 1 depicts the MPPR topological structure of the main circuit. Three ports and 2 DC Buses make up the system. PV is the system's principal power source, while Port 1 is its access port. The BES accessing port, which can modify and augment the PV output, is on port 2. Port 3 is a PPC port, which is used to manage the power provided by the voltage differential between the DC bus and the PV module, allowing MPPT and HVRT to be realised. In the system, there really are two DC buses, Bus 1 and Bus 2,
power and controls the stability of u3. A non-isolated dual half-bridge (DHB) connects Bus 1 & Bus 2 to allow for transfer control of BES and PV power. The disturbance observation approach is used for the Maximum power tracking of PV modules. In order to execute the MPPT of a PV array, the Port 3 voltage u3 must be modified in real time to guarantee that the PV array's output voltage has always been at the maximum power output, as illustrated in Figure 2.

This type of partial power conversion mode uses a non-isolated converter to achieve PPC, which can enhance energy transfer efficiency while reducing system space, weight, and cost. The proposed PPC technique in this research can also help in conducting out the HVRT in the event of a grid voltage surge. When all of the loads are linked to the DC bus via a power electronic converter in this way, the voltage of the Bus1 u_{Bus1}canfluctuatewithinacertainrange. When the grid voltage rises, the DC bus voltage uBus1 may be adjusted with the help of Port 3 to match the AC voltage. The DC bus energy uBus1 is the sum of the Port 3 voltage u3 and the PV output voltage u_{PV}, where the MPPT control determines the PV output voltage u_{PV}. On the assumption of maintaining the PV functioning at the MPPT, the Port 3 voltage u3 should be raised in order to boost its Bus 1 voltage uBus1 on the DC side of three phase converter while riding through the grid-side voltages well. In the event of a high-voltage failure, the voltage on Bus 1 u_{Bus1} should be correspondingly raised, and the voltage needed for Port 3 is$u_{3=}u_{Bus1}-u_{PV}$.

II. CONTROLSCHEMEOFTHEPOWERROUTER

Bus 1's voltage is controlled by a grid-connected three-phase converter, while Bus 2's voltage is controlled by a bi-directional dual half-bridge (DHB) DC/DC converter. Furthermore, the MPPT implementation of PV arrays necessitates the coordination of Bus 1 and Port 3. The BES unit controls out charge and discharge in accordance with the system's full energy efficiency optimization criteria.

A. VOLTAGECONTROLOFBUS1

ThecontrolschemeofBus1voltageu_{Bus1}isshowninFigure 3. Voltage and current double closed-loop control is used in the grid-connected condition. Since reference values i∗d,i∗q and feedback values id,iq are DC signals in steady state,current tracking control without steady state error can be realised through PI regulator. According the control scheme of converter, the control objective is that the voltage and power can be matched in real time to maintain the stability of both the Dc link voltage and perform the MPPT

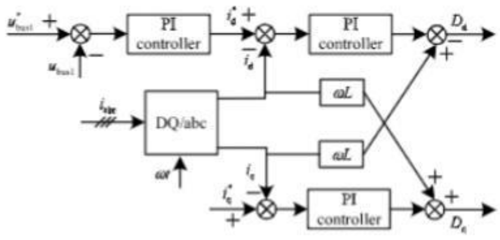

FIGURE 3. The control strategy of u_{Bus1}.

of Photovoltaic in the grid-connected condition. To suit the system's needs, the converter can operate in inverter or rectifying mode depending on the energy requirements

B. VOLTAGECONTROLOFBUS2

Figure depicts the suggested quasi DHB bi-directional inverter for PV cell applications. The circuit is made up of two half bridges, one on either side of the transformer..

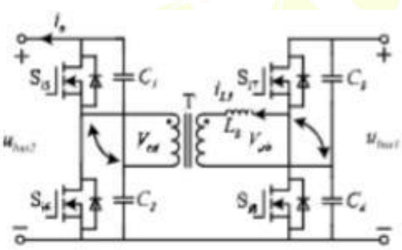

FIGURE 4. The topology of DHB bi-direction DC/DC converter.

The components of the DHB are only half the size of the full-bridge bi-directional converter. Changing the phase shift angle between the driving impulses of two half-bridge unit at both turns of the transformer, which is very easy, may regulate the power flow.

The four switches are on half-duty, the gate signal here between bridge arms is angled, and phase-shifted high-frequency voltage square waves $V_{ab}$and$V_{cd}$are created on both sides of the transformer.

The transformer's turns ratio is $N_1/N_2= u_{Bus2}/u_{Bus1}$. The power flow equation is developed from the square wave (DFC=0.5) example.

$$\frac{N_1 u_{Bus1} u_{Bus2}}{2N_2 \pi f_s L_1} \qquad \frac{|\phi|}{\pi}$$

The power flow is obviously bidirectional.

Figure shows the ideal voltage and current wave forms throughout the circuit analysis.

Figure 6 depicts the control method for the Dc link voltage uBus2. The DHB bi-directional DC-DC converter regulates the uBus2's stability. The duty cycle signals of the shifting tubes in the bidirectional converter are acquired after PI regulation and phase -shift keying control by comparing the uBus2 with its

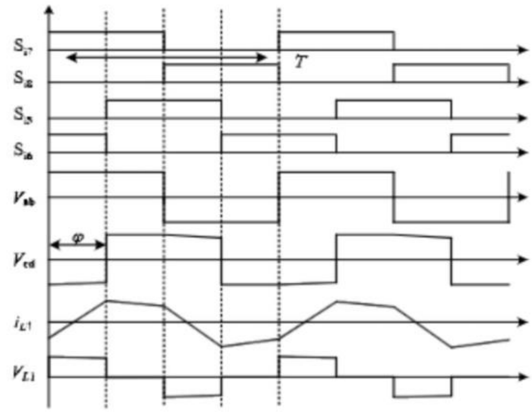

FIGURE 5. The typical waveform of DHB bi-direction DC/DC converter.

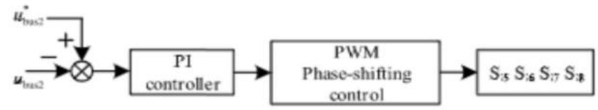

FIGURE 6. The control strategy of u_{Bus2}.

reference value. At Si5 and Si6, Si7 and Si8, the switching tubes' driving signals are complimentary. The direction of energy flow is controlled by a phase shift difference between both the bridge arms.

C. BESCHARGEANDDISCHARGECONTROL

The energy flow direction via the DHB bi-direction DC/DC converters is also impacted by the operating states of the BES in system modes, and the buck/boost converter controls the energy flow of the BES. Because the battery's SOC control method has been thoroughly investigated in [26]–[28], and is not the subject of this study, the SOC controller is not designed independently. In the control technique, the present closed-loop is used. The BES continues to balance power and energy according to energy management requirements, which are not detailed in this study. The control block diagram of the BES is shown in Figure.

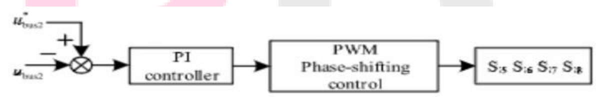

FIGURE 6. The control strategy of u_{Bus2}.

D. HVRTANDPVOPERATIONCONTROL

Controlling u_3 allows PV array MPPT to be achieved. To guarantee that now the Pv system can always complete this same MPPT in the system's energy transfer process, the Port 3 voltage u_3 in the Photovoltaic power grid-connected system must be modified in real time to ensure that the PV array's output voltage is the voltage value necessary to complete the MPPT.

The control scheme of Port 3 voltage u_3 is shown in Figure .

The stability of u_3 is controlled by bi-directional u_{Bus2}voltage, buck/boost, and buck/boost The Bus1voltageu_{Bus1}1is the sum of the Port3 voltage u3 and the PV output voltage u_{PV} in the general structure of the system, therefore the reference value is $u_3^* u_{Bus1} u_{PV}^*$. The MPPT control determines the reference value of PV output voltage u_{PV}, and the reference value of DCbusvoltageuBus1 is equal to a fixed value under typical AC voltage conditions.

Using a PI regulator and PWM link to operate the switching tubes Si1, Si2 complimentary, the duty ratio signal may be produced by comparing Port3 voltage$u_3$with its reference value. The Bus2 controls the

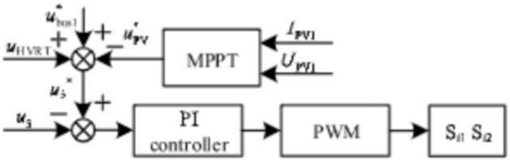

FIGURE 8. The control strategy of HVRT and PPC.

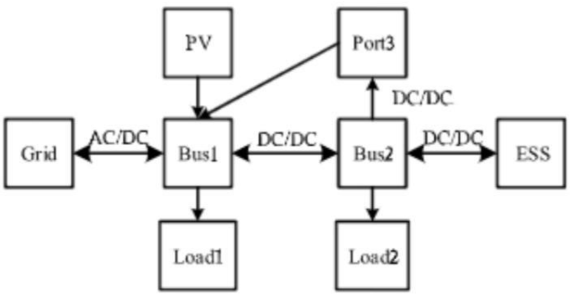

FIGURE 9. The overall energy flow diagram.

Port3 voltage $u_3$to achieve the MPPT of PV array through the clamping action of Bus1.

When a high voltage surge is detected on the AC grid, the difference value of u3 will be overlaid with an increments variable u_{HVRT} to keep the system running normally as long as the PV output stays unaltered.

When the grid voltage returns to normal ,u_{HVRT} drops to zero, and the system returns to a stable condition.

E. ANALYSISOF OVERALL ENERGY FLOW

The energy flow diagram is shown in Figure 9. PV transfersall energy to the Bus 1. Combined with the relationshipbetweenBES,PVandbusload,theenergyflowdirectionofthesystemcanbesummarizedasfollows.

When the three-phase converter is in rectifier mode,Q_{Grid} Q_{PV} Q_{Load1}>0,theenergyistransferredtoBus 2 through DHB bi-direction DC/DC converter. $Q_{Grid} Q_{PV}$ Q_{Load1}<0,energyistransferredtoBus1throughDHBbi-directionDC/DCconverter.

When the three-phase converter is in inverter mode,Q_{PV} Q_{Grid} Q_{Load1}>0,theenergyistransferredtoBus 2 through DHB bi-direction DC/DC converter. $Q_{PV} Q_{Grid} Q_{Load1}$<0, the energy is transferred to Bus 1

Figure depicts the energy flow diagram. All energy is sent to Bus 1 by PV. When the relationship between BES, PV, and busload is taken into account, the energy flow direction of the system may be summarised as follows.

$Q_{Grid} Q_{PV} Q_{Load1}$>0, The energy is transported to Bus 2 through the DHB bi-direction DC/DC converters while the three-phase converter is in rectifier

$Q_{Grid} Q_{PV} Q_{Load1}$<0,energyistransferredtoBus1throughDHBbi-directionDC/DCconverter.

The energy is transported to Bus 2 through the DHB bi-direction DC/DC converter when the three-phase converter is in inverter mode.

I. SIMULATIONANDEXPERIMENTALVERIFICATIONS

A. SIMULATIONRESULTSOFCASEI

Because the system operates in a variety of situations, different working conditions will indeed be changed in the simulation in order to validate the system's working principle. Simulation parameters are shown in Table.

The simulation time is 2 seconds, and the system operates in Case I for 0-1 seconds and Case II for 1-2 seconds.

Figures 10 and 11 illustrate the outcomes of the simulation.

Figure shows the present direction, which is positive.

TABLE 1. Parameters of the experimental prototype.

Parameters	Value
Bus 1 voltage/ $u_{Bus\,1}$	700V
Bus 2 voltage/ $u_{Bus\,2}$	400V
Port 3 voltage/ u_3	100V
BES voltage of/ u_{bes}	200V
Voltage at PV/ u_{mpp}	600V
Current at PV/ i_{mpp}	20A
Maximum power/ P_{max}	12kW
Grid voltage	380V,50Hz
Load on $u_{Bus\,2}$	100Ω
Load on $u_{Bus\,1}$	100Ω
L_1	1×10^{-4}H
L_2	5×10^{-4}H
L_3	8.5×10^{-5}H
Filter inductances/ $L_{a1}L_{b1}L_{c1}$	3.698×10^{-3}H
Filter inductance/ $L_{a2}L_{b2}L_{c2}$	1.849×10^{-3}H
Filter capacitor/ C_n	8.026×10^{-6}F

Case I: The Bus 1 voltage uBus1 is 700V before and after switching operating circumstances, as indicated in Figure 10. Bus 2 voltage u Bus2 is 400V. The Port 3 voltage u3 is 100V.The BES is charged with a continuous current of 30A during 0-1s and a flow of 30A during 1-2s.The PV module's output current is constant at 20A. It can be observed from the direction of flow of inductive current iL1 that regardless of the operating situation, energy is constantly travelling via the buck/boost circuit to capacitor CP3 to raise Ports 3 voltage u3. After calculations, the partial power conversion value is 2000 W, or around 1/6 of PV capacity.The grid current and voltage are at the same frequency and phase during 0-1s. The three-phase converter is in rectifier mode at this moment, and the power grid is transferring energy to Bus1.

The output curves of the phase-shiftingcontrol module's regulator, as well as the steady-state waveform of the voltage and current of the phase-shifting inductanceL3, are displayed.

Figure illustrates this. The regulator's output is positive, and the DHB bi-direction DC/DC converter is operating in forward mode. Through the DHB bi-direction DC/DC converter, the energy is delivered to

Bus 2. The three-phase converter operates in inverter mode for 1-2 seconds. Bus 1 is responsible for transferring energy to the grid. The output of the phase shift control module's controller is negative, and the DHB

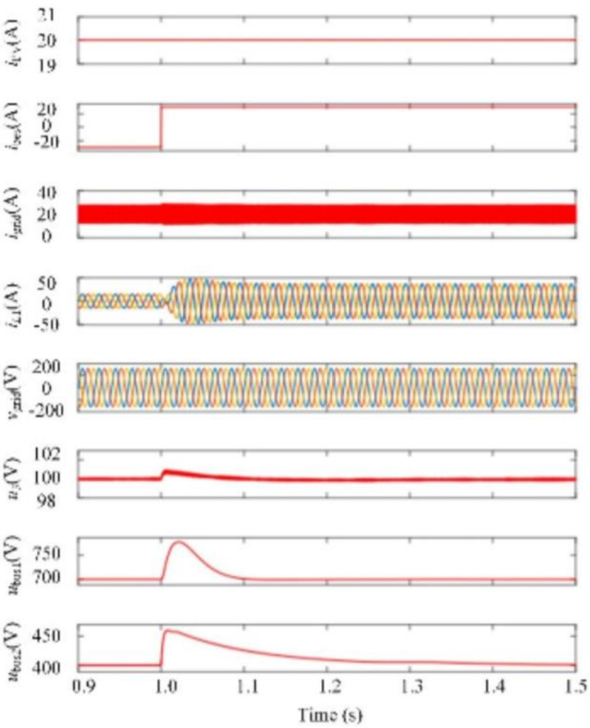

FIGURE 10. Simulation result of case I.

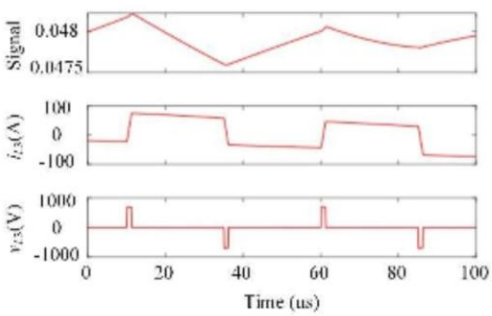

FIGURE 11. The steady-state waveform in 0-1s.

bi-direction DC/DC converter is in reverse mode. Through the DHB bi-direction DC/DC converter, the energy is transmitted to the Bus1. Inductance's current and voltage waveform both attain a steady state..

B. EXPERIMENTRESULTS

Circuit rated parameters areas follows:Bus 1voltage /u_{Bus1}: 100V;
Bus 2voltage/u_{Bus2}:50V;Auxiliaryport voltage/u_3:0xV;
PV port voltage/ u_{PV}: ~(100-x)100V;PV output power/ P_{PV}: 250W;Transformerratio:2/1.

In order to further verify the effectiveness of the proposed scheme, a down-scaled experimental platform as shown in Figure was built in research. The experimental result of partial power conversion is shown in Figure. The Port3 voltage u3 is adjusted in the circuit loop to implement the MPPT of the PV module. The PV power curve varies dramatically as u3 rises. However, there is only one point with the highest PV power output in the Figure..

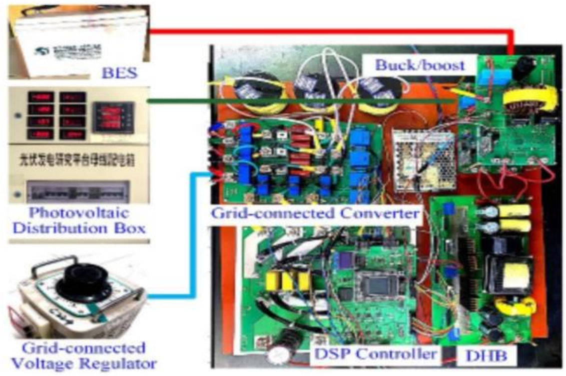

FIGURE 14. Experimental platform.

The partial power figure is 47.1W, which is around 21% of the PV capacity. Partial power is transmitted to the capacitor CP 3 side to modify the Terminal 3 voltage u3 so that the PV module always tracks the voltage output at maximum power, completing the MPPT control of PV grid-connected Photovoltaic cells system. The waveform of the principal buses & ports are displayed in Figure while the system is operating at maximum power. Figure shows the

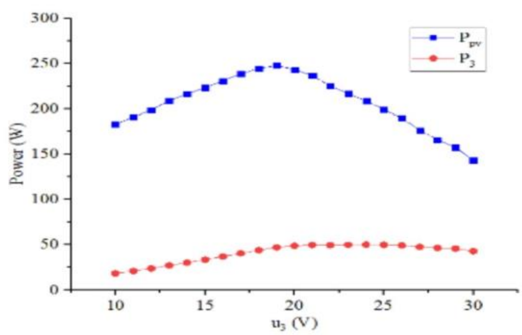

FIGURE 15. Experiment result of partial power conversion of PV.

functioning by the bi directional buck/boost will vary As a

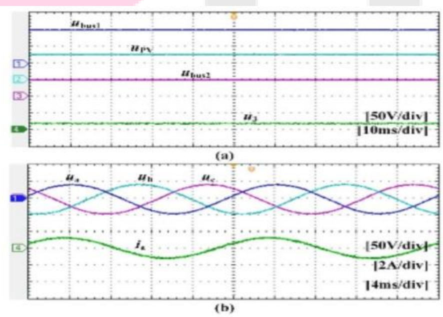

FIGURE 16. Experiment waveforms when system operating at MPPT. (a) Voltage of Bus 1, PV, Bus 2 and Port 3. (b) Voltage and current of AC side.

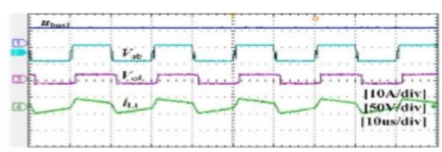

FIGURE 17. Operating waveforms of DHB.

result, the total voltage of Load buses will be increased but the PV output will stay unaltered, allowing the system to function correctly in the event of a high-voltage malfunction and completing the HVRT.
Graph. Figures demonstrate the energy delivery responses waveforms of the BES whenever the Photovoltaic grid-

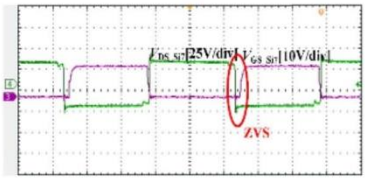

FIGURE 18. The ZVS waveforms of DHB.

connected power changes (b). The reaction of the DHB is illustrated in Figure 20(c) while the system is running at HVRT, and the power produced by the DHB stays constant,

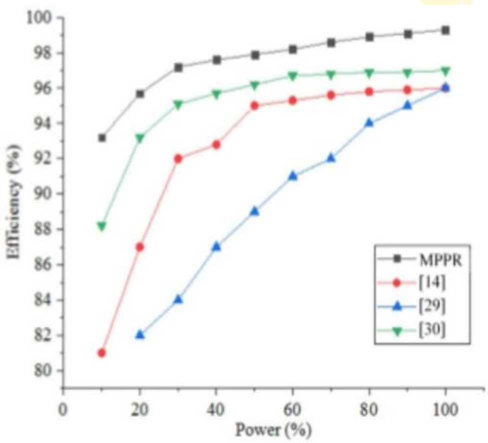

FIGURE 21. Efficiency comparison result of DC/DC stage with other full power conversion schemes.

suggesting that PV power production units can maintain a consistent output throughout HVRT. The switching device's on-state resistance and transient response parameters are to blame. PPC has a greater overall efficiency than FPC due to the lower power of the converter handling. It's also worth mentioning that when PV output hits 30% of rated power, PPC conversion efficiency does not very much, which is critical for PV systems with a broad range of output power variations.

II. CONCLUSION

This article suggests a multi-port power routing system based on PV batteries. In comparison to the standard PV-battery grid-connected system, the proposed MPPR in this paper has two primary features accomplished concurrently by one auxiliary port: the first one is the power converter conversions of the DC/DC stage, which greatly enhances powertransferefficiency. Secondary,MPPRrealizesHVRTonthepremiseofmaintainingnormalPVoutput,andauxiliaryportadapts to grid-side voltage swells by altering its voltage in order to increase the voltage level of the three phase converter DC bus. The system can flexibly realize the power exchange between three ports, two DC buses and the grid.

REFERENCES

[1] A.Sangwongwanich,Y.Yang,andF.Blaabjerg,‘‘High-performancecon-stantpowergenerationingrid-onnectedPVsystems,’’*IEEETrans.PowerElectron.*, vol. 31, no.3, pp. 1822–1825, Mar.2016.

[2] C.Zhong,Y.Zhou,X.Zhang,andG.Yan,‘‘Flexiblepower-point-tracking-based frequency regulation strategy for PV system,’’ *IET Renew. PowerGener.*, vol. 14, no. 10, pp.1797–1807, Jul. 2020.

[3] H. Fathead, ‘‘Improving the power efficiency of a PV power generation system using a proposed electrochemical heat engine embedded in the system,’’ *IEEE Trans. Power Electron.*, vol. 34, no. 9, pp. 8626–8633,Sep. 2019.

[4] Y.Liu,S.You,andY.Liu,‘‘StudyofwindandPVfrequencycontrolin U.S. power grids—EI and TI case studies,’’ *IEEE Power Energy Technol.Syst. J.*, vol. 4,no. 3, pp.65–73, Sep. 2017.

[5] H.Sugihara,K.Yokoyama,O.Saeki,K.Tsuji,andT.Funaki,‘‘Economicand efficient voltage management using customer-owned energy storagesystems in a distribution network with high penetration of photovoltaicsystems,’’*IEEETrans.PowerSyst.*,vol.28,no.1,pp.102–111,Feb.2013.

[6] Y.Cai,W.Tang,O.Xu,andL.Zhang,‘‘ReviewofvoltagecontrolresearchinLVdistributionnetworkwithhighproportionofresidentialPVs,’’(in Chinese), *PowerSyst. Technol.*,vol. 42, no.1, pp.220–229, 2018.

[7] J. Xu, N. Liu, L. Yu, J. Y. Lei, and J. H. Zhang, ‘‘Optimal allocation ofenergy storage system of PV microgrid for industries considering impor-tant load,’’ (in Chinese), *Power Syst. Protection Control*, vol. 44, no. 9,pp. 29–37, 2016.